

1 P4 C1 — Three disjoint congruence classes

Solved — complete written proof. For any three pairwise disjoint integer congruence classes with positive moduli, some pair of moduli has greatest common divisor at least 3.

For an integer a and a positive integer m , write

$$a \pmod{m} = \{a + mt : t \in \mathbb{Z}\}.$$

Repeated moduli and residues are allowed. We first establish the precise intersection test used below. Let $a \pmod{m}$ and $b \pmod{n}$ be two classes, and set $g = \gcd(m, n)$. They intersect if and only if g divides $b - a$. Indeed, if an integer z lies in both classes, then $m \mid z - a$ and $n \mid z - b$. Since g divides both m and n , it divides

$$(z - a) - (z - b) = b - a.$$

Conversely, suppose $b - a = gt$. Bézout's identity gives integers x, y with $g = mx + ny$. Set $z = a + mxt$. Then $z - a = mxt$ is divisible by m , while

$$z - b = mxt - (b - a) = mxt - (mx + ny)t = -nyt$$

is divisible by n . Hence z lies in both classes. This proves the intersection test, including when the moduli are equal.

Now let $a_0 \pmod{m_0}$, $a_1 \pmod{m_1}$, and $a_2 \pmod{m_2}$ be pairwise disjoint. Assume, toward a contradiction, that every pair of moduli has gcd less than 3. Every such gcd is positive. It cannot equal 1: by the intersection test, a pair with coprime moduli would intersect because 1 divides every integer. Therefore

$$\gcd(m_0, m_1) = \gcd(m_0, m_2) = \gcd(m_1, m_2) = 2.$$

For each pair $i \neq j$, disjointness and the intersection test imply $2 \nmid a_j - a_i$. Thus a_0, a_1, a_2 are pairwise distinct modulo 2. There are only two residue classes modulo 2, so this is impossible. Consequently at least one pair satisfies $\gcd(m_i, m_j) \geq 3$, as required.

The bound is attained: $0 \pmod{3}$, $1 \pmod{3}$, and $2 \pmod{3}$ are pairwise disjoint, and every pair of their moduli has gcd exactly 3.

1.1 Formal verification

The written argument above is complete independently of software. It is also formalized as `ProofPursuit.P4.C1.solution : Statement 3` in [C1.lean](#). [Basic.lean](#) defines actual set-theoretic disjointness and proves the needed Bézout/CRT implication. The project pins Lean 4.34.1 and uses bundled `Std`. From `cagent/p4/lean`, the focused commands are:

```
lake build ProofPursuit.P4.C1
lake env leanchecker ProofPursuit.P4.C1
```

The [verification record](#) reports a successful build and independent kernel check of the containing C3 package, including C1, with no unproved custom axioms or `sorry`.